16 WEEK PLAN / COURSE DETAILS

By

DEPARTMENT OF PYSICS

UNIVERSITY OF GUJRAT

***Course Code**: GE SE-102*	***Course Title:** Applied Physics*
***Credit Hours**: 03*	***Maximum Lectures:** 32 (1.5hours each)*
***Course Type**: Minor*	***Pre-requisites**: Applied Physics and electromagnetic theory*
***Year**: Fall 2023*	***Semester**: 1st*
***Instructor's Name**: Ms. Falak Shafiq*	***Office (Room No)**: S-112, Physics Dpt. S-Block,Hafiz Hayat Campus, UOG*
***E-mail**: 8089@uog.edu.pk*	***Students' Contact house**: Studentsare welcomed any time during officehours to visit instructor's office; however, sixty minutes each after everyclass are reserved for the students of this class.*
*C**lass Level**: BS-1st*	***Department:** Software Engineering*

CLASSES REGISTRATION:

Following students of MSc/BS will be enrolled for this Course

S#	*R. No.*	*Name of Student*	*Email addresses*	*Cell No.*
1				

ATTENDANCE POLICY:

- *As per University policy, a minimum of 70% attendance shall be required for a studentto be eligible to appear in the Mid and End Term examinations. All students concernedare advised to keep in touch with their attendance records through their UOG email account.*
- *The students with less than 70% attendance shall not be allowed to take classes after Mid Term Examination dates.*

CLASS RULES AND REGULATIONS:

- *Without registration and UOG ID card no student shall be allowed to attend the class. Students are advised to display the UOG card in a visible position.*
- *The class shall be started as per given time table. No one shall be allowed after 10Minutes of the commencement of class unless genuine excuse.*
- *Students are encouraged to interrupt by asking the questions during the lecturer.*

- *Drinking, smoking, mobile usage, sectarian or political discussion and immoral /unethical activities shall not be allowed during class.*
- *Students are welcome any time during office hours to visit instructor's office.*
- *All the communication regarding any lecturer or change of class etc. (If any) shall be forwarded directly to the students' given email inbox; however, the hard copies for Xerox (if any)shall be given to any responsible student to place at the photo copier shop. Thus students advised to keep in touch with their-box.*

TEACHING & EVALUATION METHODOLOGY:

CLASS LECTURER:

- *English shall be the language of instructions; however, shy or scrawny students shall be allowed to communicate in Urdu.*
- *Each lecture shall be of 1.5 hours.*
- *Audio video aids like Computer(s), Multimedia and White Board shall be used asteaching tools.*

STUDENT'S EVALUATION (GRADING):

*The course shall be evaluated out of **100 Marks** as per following division:*

GRADING DETAIL:

Sessional Work (25Marks):

• *Quiz:*	*12 marks*
• *Presentation/Project*	*07 marks*
• *Assignments*	*06 marks*
Total:	***25 Marks***

Mid Term Examinations (25 Marks)*:*

- *Short and long question type questions*

End Term Examination (50 Marks);

- *Long, short questions and Problems*

Course Outline

Applied Physics

Electric force and its applications and related problems, conservation of charge, charge quantization, Electric fields due to point charge and lines of force. Ring of charge, Disk of charge, A point charge in an electric field, Dipole in a n electric field, The flux of vector field, The flux of electric field, Gauss' Law, Application of Gauss' Law, Spherically symmetric charge distribution, A charge isolated conductor, Electric potential energy, Electric potentials, Calculating the potential from the field and related problem Potential due to point and continuous charge distribution, Potential due to dipole, equipotential surfaces, Calculating the field from the potential, Electric current, Current density,

Resistance, Resistivity and conductivity, Ohm's law and its applications, The Hall effect, The magnetic force on a current, The Biot- Savart law, Line of B, Two parallel conductors, Amperes' s Law, Solenoid, Toroid's, Faraday's experiments, Faraday's Law of Induction, Lenz's law, Motional emf, Induced electric field, Induced electric fields, The basic equation of electromagnetism, Induced Magnetic field, The displacement current, Reflection and Refraction of light waves, Total internal reflection, Two source interference, Double Slit interference, related problems, Interference from thin films, Diffraction and the wave theory, related problems, Single-Slit Diffraction, related problems, Polarization of electromagnetic waves, Polarizing sheets, related problems.

Recommended Books

1. *Fundamentals of Physics (Extended), 10th edition, Resnick and Walker*
2. *Narciso Garcia, Arthur Damask, Steven Schwarz., "Physics for Computer Science Students", Springer Verlag, 1998*

EXPECTEDLEARNINGOUTCOMES:

The students shall be able to have expertise in applied physics in the research domain.

SCHEDULE:

Course Planning for Semester Fall-2023

Week	Lectures	Topics	Sub-Heading/-Topic **Note:** Sub-topic related examples areal so part of it.	Page No.	Numerical Problems	Book
1	1-2	**Coulomb's Law:**	Conservation of charge, Charge quantization, Electric force and its application and related problems	613-623		Fundamentals of physics
2	3-4	**Electric Fields:**	Electric field due to point charge and lines of force, Ring of charge, Disk of charge, A point charge in an electric field, Dipole in an electric field	633-650		Same as above
3	5-6	**Gauss's law:**	A charged isolated conductor, The flux of vector field, The flux of electric field, Gauss's law,	659-670		Same as above
4	7-8	**Gauss's law:**	Application of Gauss's law, spherically symmetric charge distribution	671-677		Same as above
5	9-10	**Electric Potential:**	Electric potential energy, Electric potentials	685-690		Same as above
6	11-12	**Electric Potentials:**	Equipotential surfaces, Calculating potential from field	690-693		Same as above
7	*13-14*	***Electric Potentials:***	*Potential due to an electric dipole, Potential due to charged particle*	*694-697*		*Same as above*
8	*15-16*	***Electric Potentials:***	*Potential due to point and continuous charge distribution, calculating field from potential*	*698-702*		*Same as above*
9	***Mid Term Exams***					
10	*17-18*	***Current and resistance:***	*Electric current, Current density, Resistance, Resistivity and conductivity*	*745-754*		*Same as above*
11	*19-20*	***Current and resistance:***	*Ohm's law and its application,*	*756-759*		*Same as above*
12	*21-22*	***Magnetic Fields:***	*Hall Effect, The Biot-Savart Law, Line of B, Two Parallel Conductors,*	*803-815*		*Same as above*
13	*23-24*	***Magnetic Field due to Current & Inductance and Induction:***	*The magnetic force on a current, Ampere's law, Solenoid, Toroid, Faraday's experiments, Faraday's law*	*836-851*		*Same as above*

			of induction, Lenz's Law, Motional EMF, Induced electric field	*864-876*		
14	*25-26*	***Maxwell's equation and magnetism matter***	*The basic equation of electromagnetism, induced magnetic field, The displacement current*	*941-946*		*Same as above*
15	*27-28*	***Electromagnetic wave &Interference:***	*Polarization, Reflection and refraction of light waves, Total internal reflection, two source interference, Double slit interference, Interference from thin films*	*985-996* *1053-1065*		*Same as above*
16	*29-30*	***Diffraction:***	*Diffraction and wave theory, Single slit Diffraction, Polarization of electromagnetic waves, polarizing sheets and related problems*	*1081-1088*		*Same as above*
17	*31-32*	***Project/ presentation discussion***				

Class wise Activities of the Semester-Fall-2023-PHYS-117																	
Date																	
Day/Week	***1.***	***2.***	***3.***	***4.***	***5.***	***6.***	***7.***	***8.***	***9.***	***10.***	***11.***	***12.***	***13.***	***14.***	***15.***	***16.***	***17.***
Monday									***Mid term***								
Tuesday																	
Wednesday				***A-1***							***A-2***						
Thursday																	
Friday						***Q-1***								***Q-2***			